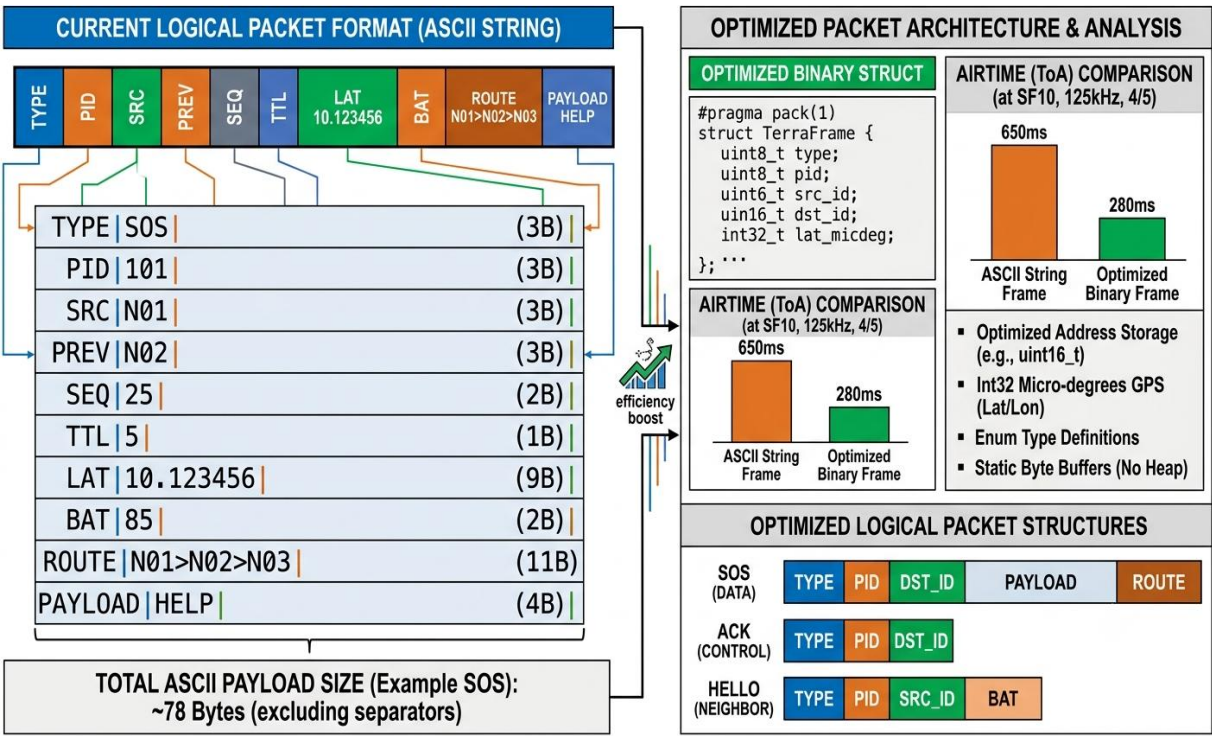


COMMUNICATION PROTOCOL

The Dynamic TerraLink Communication Protocol defines how an SOS packet is **created, routed, validated, forwarded, acknowledged, and recovered when a communication link fails.** It uses lightweight AODV-inspired routing, duplicate filtering, TTL-based loop prevention, hop-by-hop ACK, retransmission, and self-healing.

1. Packet Creation and Packet Structure

When an End Node detects an SOS event, it creates a structured packet containing the source, destination, sequence number, TTL, location, battery status, and emergency payload.



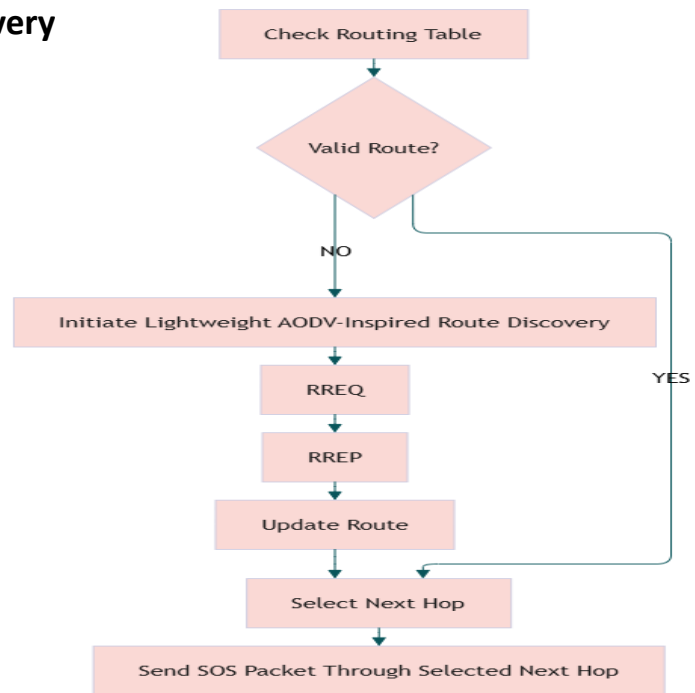
The main fields are:

- **TYPE** – Identifies the packet type.
- **PID** – Packet identifier.
- **SRC** – Source node.
- **DST** – Destination/base station.
- **SEQ** – Sequence number used for packet identification and duplicate filtering.
- **TTL** – Limits packet lifetime.
- **GPS** – Emergency location.
- **BATTERY** – Source-node battery status.
- **PAYLOAD** – Emergency information.

2. Route Selection and Route Discovery

Before transmitting an SOS packet, the node checks its routing table.

TerraLink uses a **lightweight AODV-inspired route discovery mechanism** rather than claiming full standard AODV implementation.

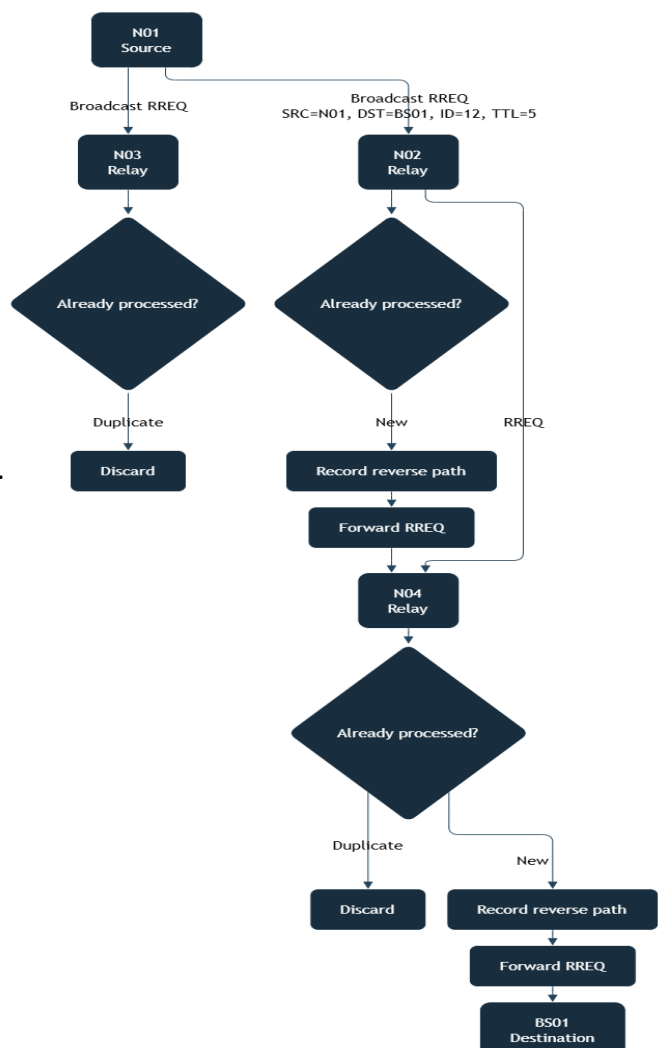


3. RREQ — Route Request

When there is no valid route, the source broadcasts an **RREQ**.

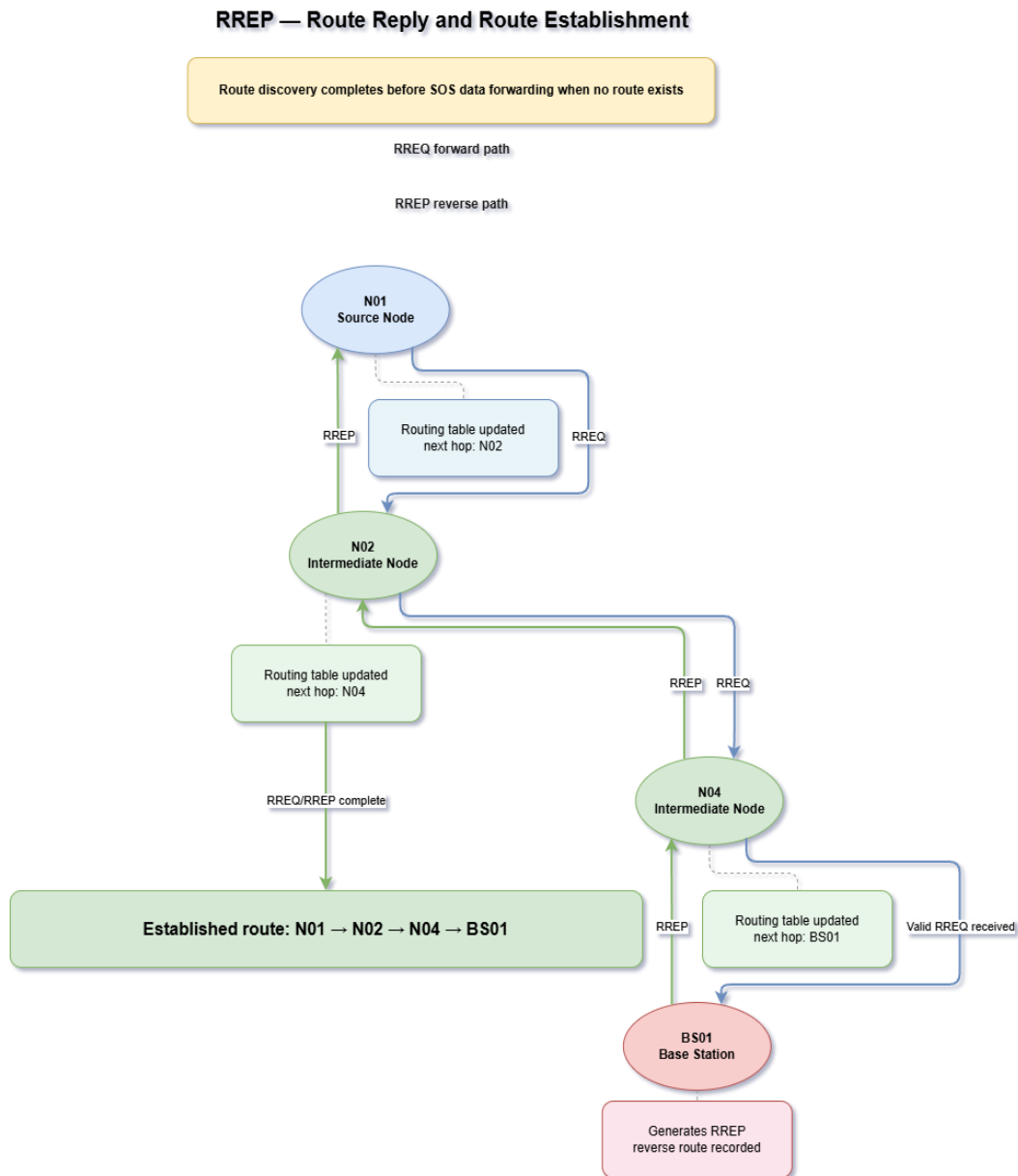
A relay receiving the RREQ:

1. Checks whether this RREQ has already been processed.
2. If duplicate → discard.
3. If new → record the reverse path.
4. Forward the RREQ toward the destination.



4. RREP — Route Reply

When the Base Station receives a valid RREQ, it generates an RREP. The RREP travels back through the reverse path:



The important point is that **RREQ/RREP establishes the route before the SOS data is forwarded when no route exists.**

5. Duplicate Filtering

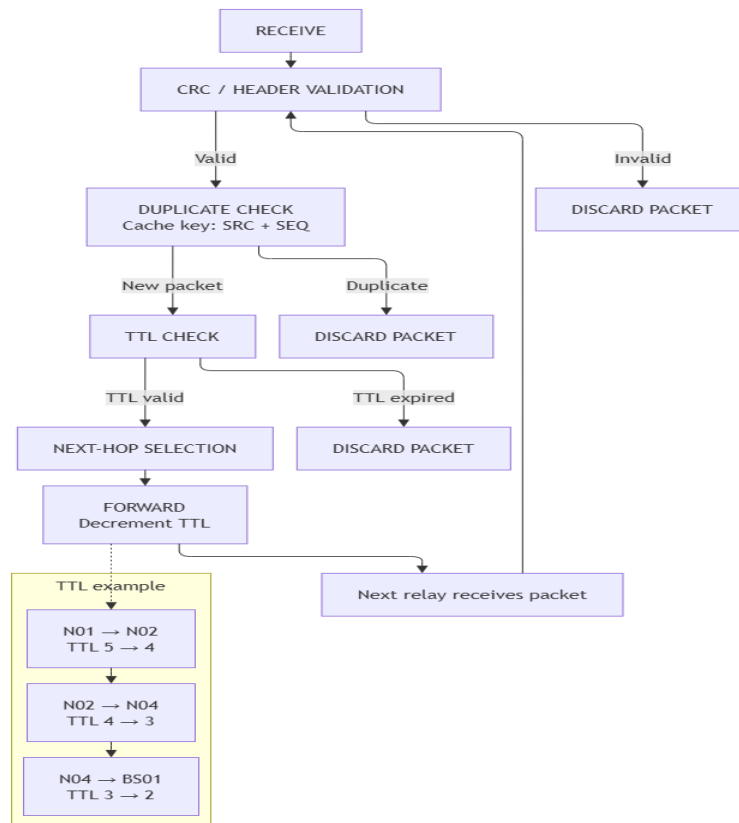
Each node maintains a small SRAM cache using:

SRC + SEQ

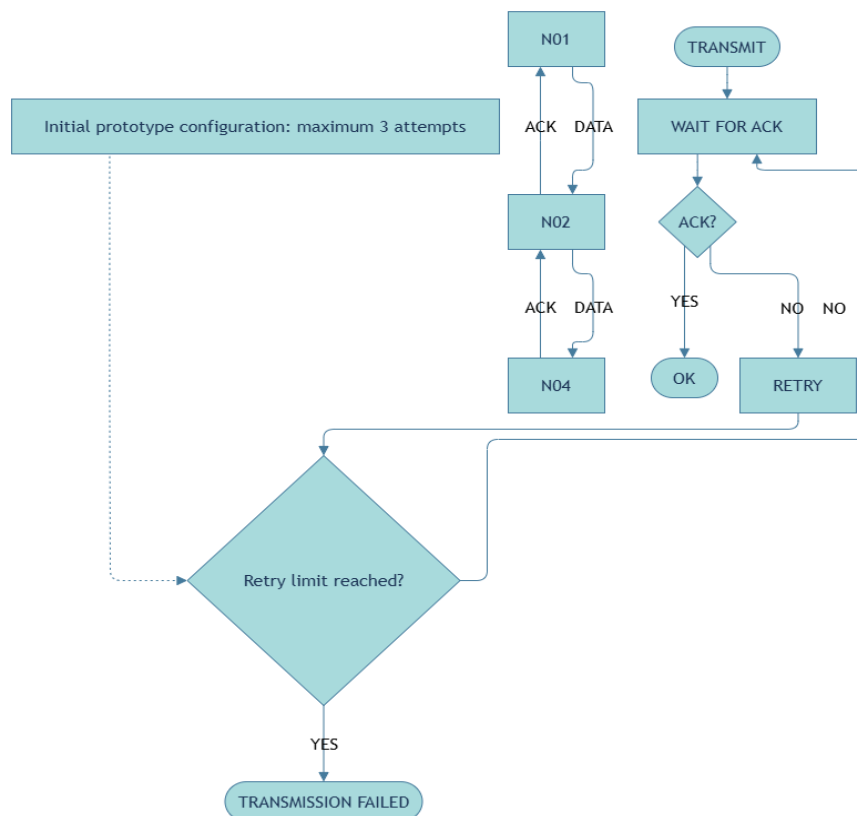
If the same source/sequence combination has already been processed, the packet is discarded to prevent repeated forwarding.

6. Packet Validation and Forwarding

Every relay validates a received packet before forwarding it. The processing order is:



7. Hop-by-Hop ACK and Retransmission

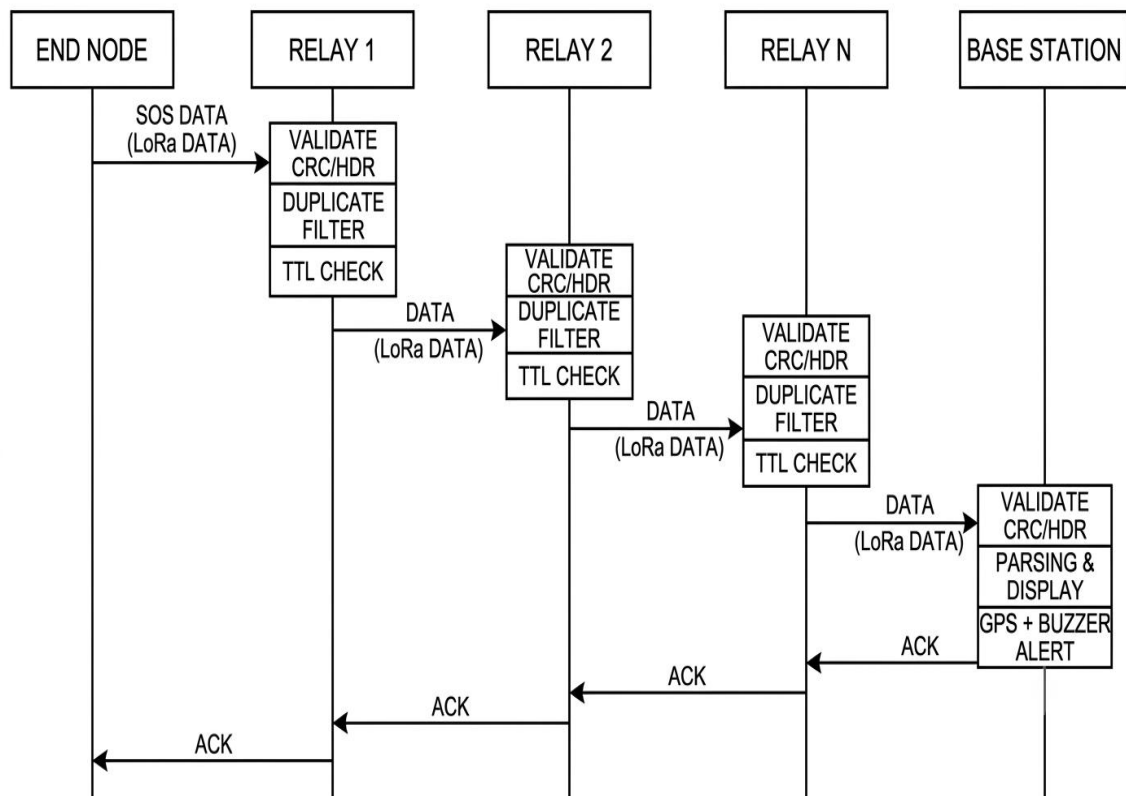
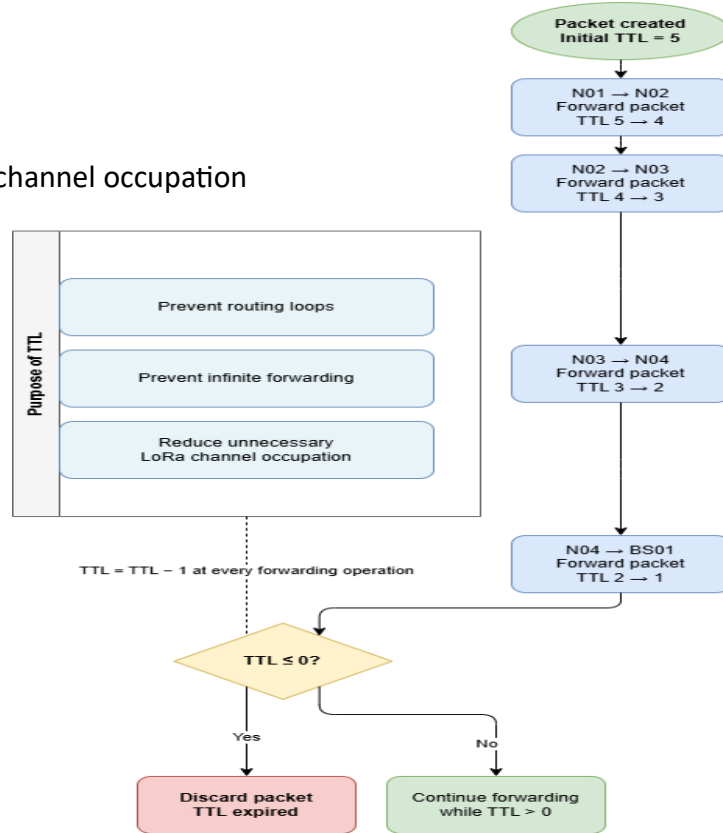


8. TTL — Packet Lifetime

TTL limits how many forwarding operations a packet can undergo.

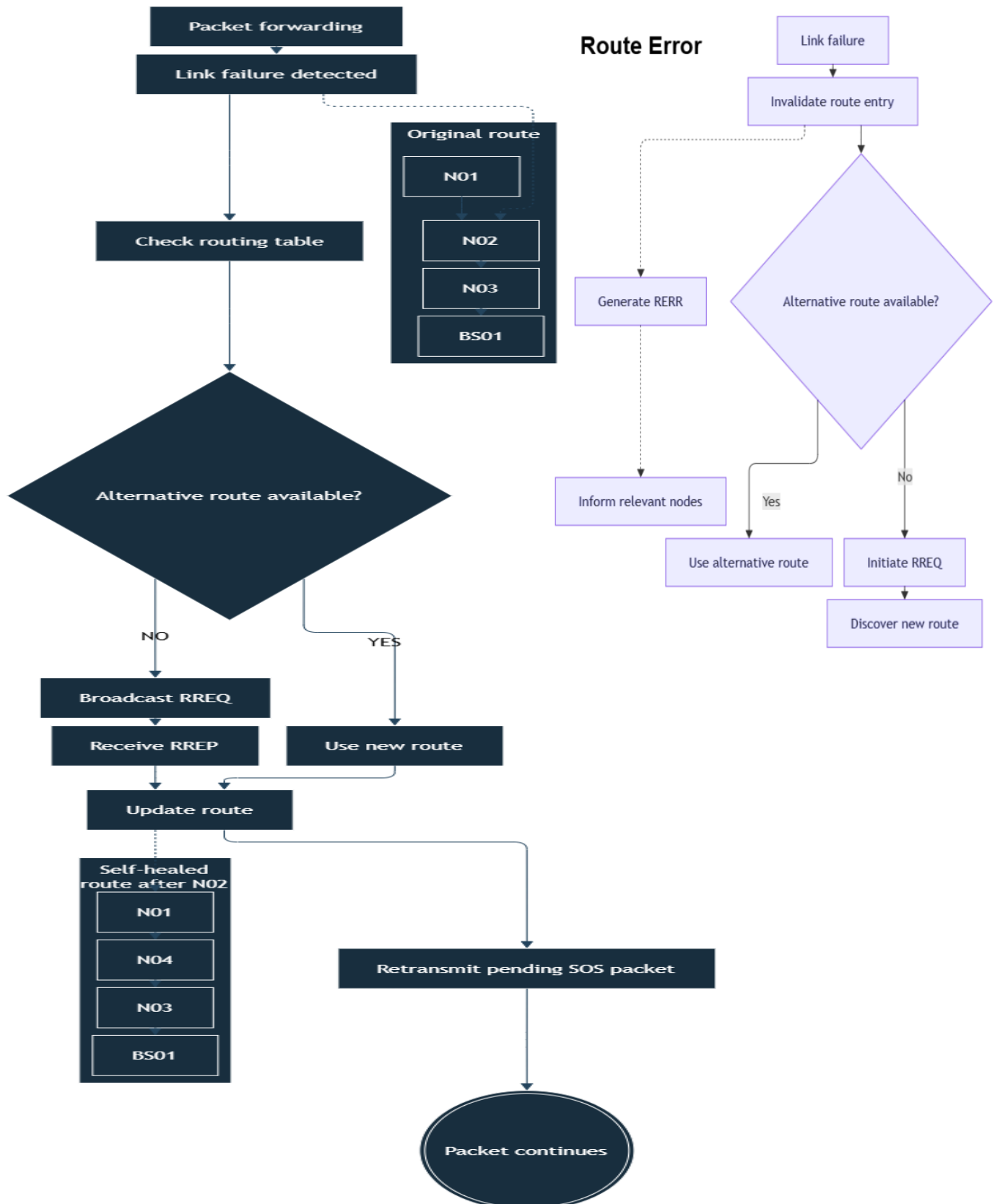
The purpose is to prevent:

- Routing loops
- Infinite forwarding
- Unnecessary LoRa channel occupation



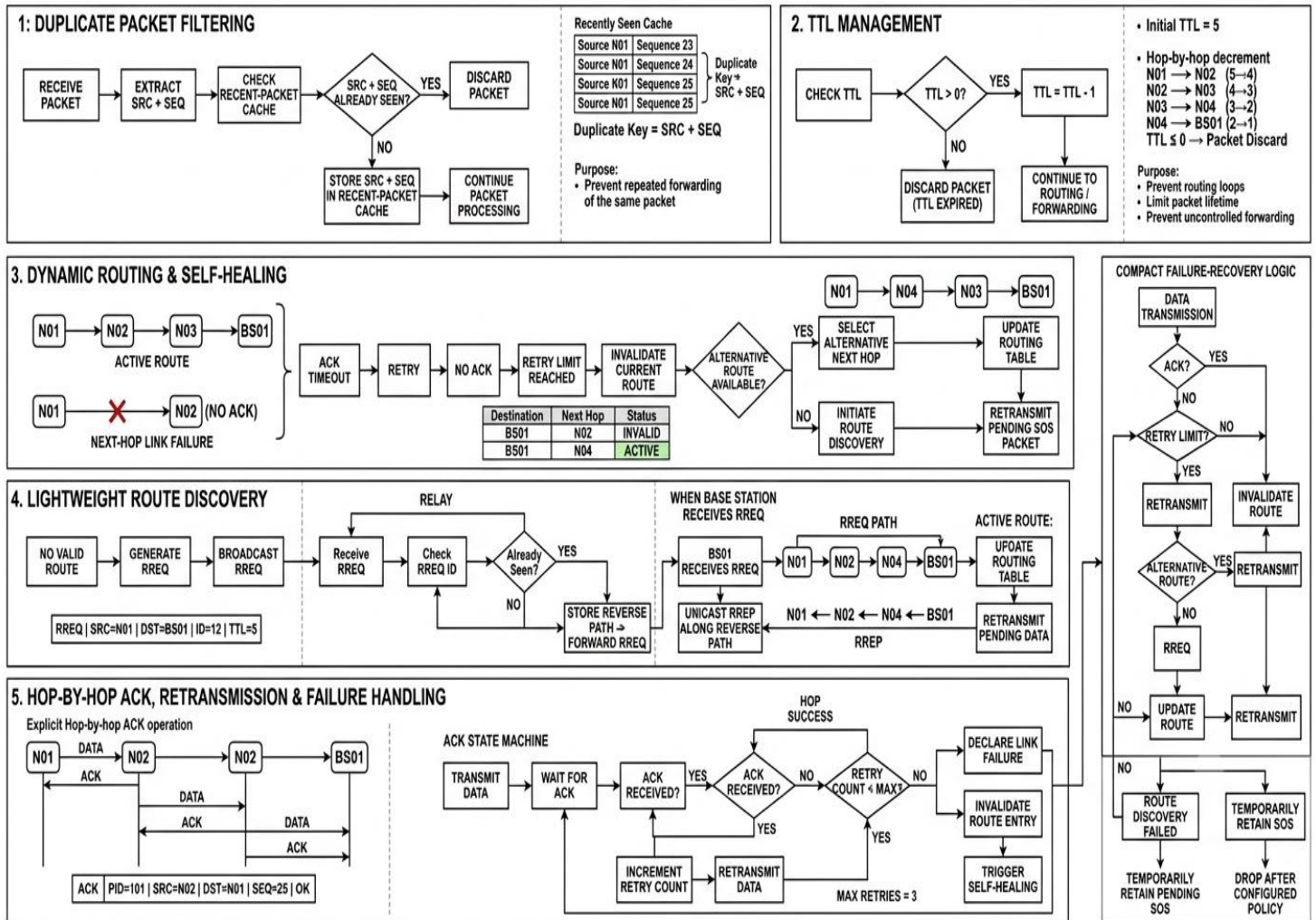
9. Self-Healing / Alternative Route

After detecting a failed next hop, the node checks whether another route is available.



DYNAMIC TERRALINK — COMMUNICATION PROTOCOL: ROUTING, TTL, ACK & SELF-HEALING

LoRa-Based Multi-Hop Emergency Communication



The TerraLink protocol has **five core mechanisms**:

- 1. Routing:** Find and select the next hop.
- 2. Validation:** Reject corrupted, duplicate, or expired packets.
- 3. Forwarding:** Move the packet hop-by-hop toward the Base Station.
- 4. Reliability:** Use ACK + limited retransmission.
- 5. Self-healing:** Detect failed links, discover an alternative route, update the routing table, and retransmit the pending SOS.

TerraLink combines **LoRa multi-hop forwarding**, **SRC+SEQ duplicate filtering**, **TTL-based loop prevention**, **hop-by-hop ACK/retransmission**, and **lightweight AODV-inspired route recovery** to provide resilient emergency-message delivery.